Logan DeFour

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Education

University of Michigan

Bachelor of Science in Engineering, Electrical Engineering

- Expected May 2028
- Relevant Coursework: Control Systems Analysis and Design, Signals and Systems, Introduction to Programming and Data Structures, Intro to Computer Organization, Introduction to Logic Design

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Experience

Devereaux Sawmill | Engineering Intern

Summer 2026

- Shadowed project manager on industrial automation and facility improvement projects
- Assisted with PLC system design
- Observed industrial troubleshooting, panel design practices, and field installation procedures

Manitou Island Transit | Deckhand & Tour Guide

Summer 2024, Summer 2025

- Delivered guided island tours to groups of 10-30 visitors
- Assisted in safe loading and unloading of passengers and cargo
- Managed beverage service and point-of-sale transactions for trips serving over 100 guests
- Worked in a fast-paced environment requiring communication, safety awareness, and customer service

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Projects

PLC Elevator Controller Simulation

- Developed a multi-routine PLC application in Rockwell CCW to control a simulated four-floor elevator using ladder logic and state-machine sequencing.
- Implemented smart-ordering, door operation, floor tracking, and sensor-driven state transitions through modular control and simulation subsystems.
- Created a virtual plant model that simulated elevator motion and floor sensors, allowing closed-loop validation of control logic in a software-only environment.

Autonomous Dead-Reckoning Robot Cart

- Designed and built an ESP32-based differential drive robot using wheel encoders, IMU sensing, and motor feedback
- Developed odometry and heading estimation algorithms for autonomous navigation
- Implemented PID control loops for motor speed and heading regulation
- Integrated sensors, motor drivers, power electronics, and embedded software into a complete robotic platform

Drone Attitude Simulation and State Estimation

- Developed a Python simulation of rotational drone dynamics with sensor noise and external disturbances
- Implemented PID-based stabilization for attitude control
- Evaluated filtering techniques including moving average, alpha-beta-gamma, and Kalman filtering approaches
- Visualized system response and controller performance using MATLAB and Python

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Skills

Programming: C++, MATLAB, Python, Verilog

Controls & Robotics: PID Control, State Estimation, Odometry, Sensor Fusion, System Modeling

Embedded & Hardware: ESP32, Microcontrollers, Sensor Integration, Motor Drivers, PCB Soldering, Electrical Wiring

Tools: VS Code, MATLAB, Rockwell CCW